

Cloud & Data Engineering - Introduction

IA Institut - 2025/2026

Section 0

Internship

Section 1

Real World Settings

A DAY IN DATA

The exponential growth of data is undisputed, but the numbers behind this explosion – fuelled by internet of things and the use of connected devices – are hard to comprehend, particularly when looked at in the context of one day

500m

tweets are sent every day

Twitter



4PB

of data created by Facebook, including

350m photos

100m hours of video watch time

Facebook Research

294bn

billion emails are sent

Radicati Group

320bn

emails to be sent each day by 2021

306bn

emails to be sent each day by 2020

3.9bn

people use emails

4TB

of data produced by a connected car

Intel

ACCUMULATED DIGITAL UNIVERSE OF DATA

4.4ZB

44ZB

PwC

2013

2020

DEMYSIFYING DATA UNITS

From the more familiar 'bit' or 'megabyte', larger units of measurement are more frequently being used to explain the masses of data

Unit	Value	Size
b bit	0 or 1	1/8 of a byte
B byte	8 bits	1 byte
KB kilobyte	1,000 bytes	1,000 bytes
MB megabyte	1,000 ² bytes	1,000,000 bytes
GB gigabyte	1,000 ³ bytes	1,000,000,000 bytes
TB terabyte	1,000 ⁴ bytes	1,000,000,000,000 bytes
PB petabyte	1,000 ⁵ bytes	1,000,000,000,000,000 bytes
EB exabyte	1,000 ⁶ bytes	1,000,000,000,000,000,000 bytes
ZB zettabyte	1,000 ⁷ bytes	1,000,000,000,000,000,000,000 bytes
YB yottabyte	1,000 ⁸ bytes	1,000,000,000,000,000,000,000,000 bytes

*A lowercase "b" is used as an abbreviation for bits, while an uppercase "B" represents bytes.

65bn

messages sent over WhatsApp and two billion minutes of voice and video calls made

Facebook



Searches made a day

5bn

Searches made a day from Google

3.5bn

Smart Insights



463EB

of data will be created every day by 2025

ioe

95m

photos and videos are shared on Instagram

Instagram Business



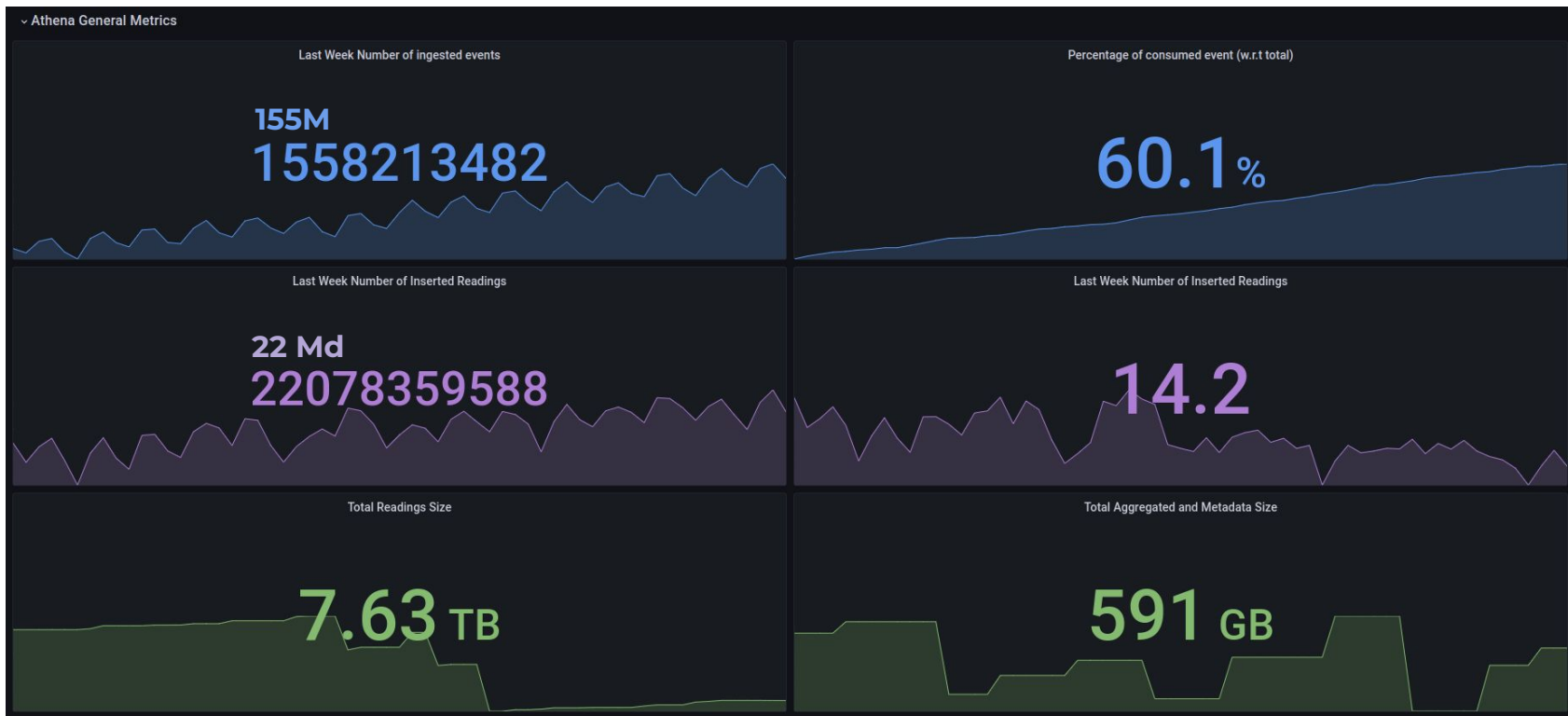
28PB

to be generated from wearable devices by 2020

Statista



Munic Use Case



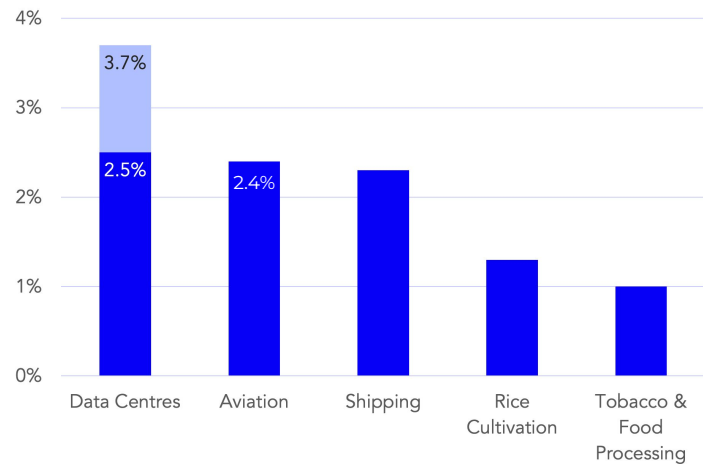


2,000km



Global cloud computing emissions exceed those from commercial aviation

Share of global CO₂ emission generated by sector/category



Source: Climatiq Analysis, The Shift Project, OurWorldinData



<https://www.climatiq.io/blog/measure-greenhouse-gas-emissions-carbon-data-centres-cloud-computing> (2022)

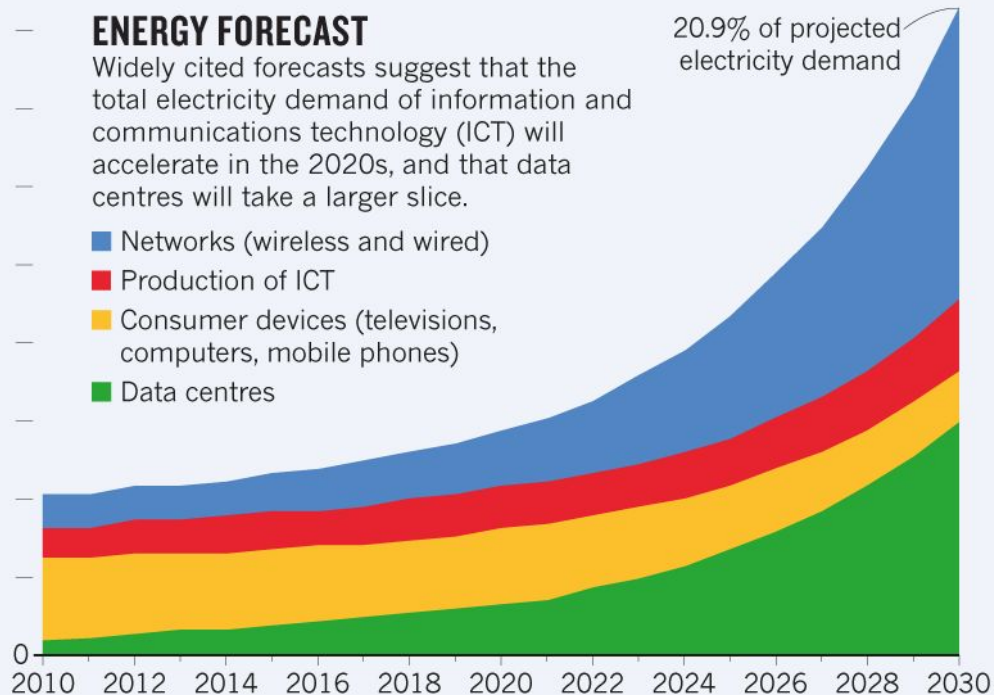
* <https://theshiftproject.org/en/article/lean-ict-our-new-report/>

9,000 terawatt hours (TWh)

ENERGY FORECAST

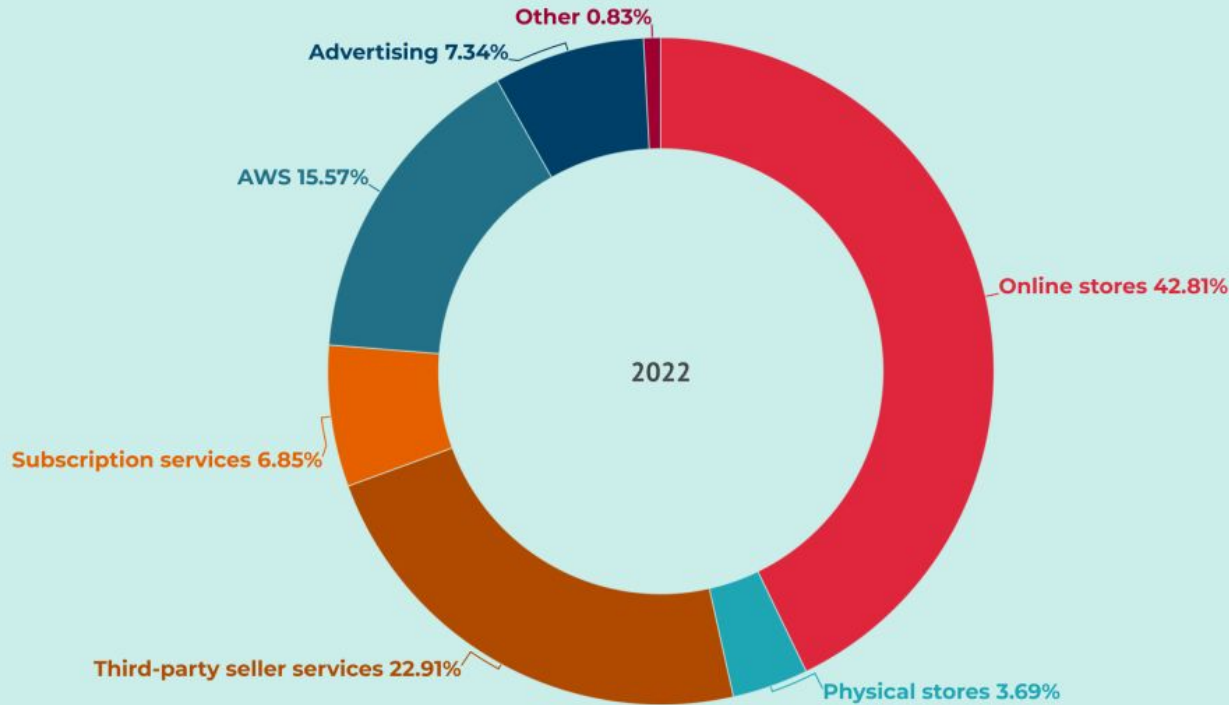
Widely cited forecasts suggest that the total electricity demand of information and communications technology (ICT) will accelerate in the 2020s, and that data centres will take a larger slice.

- Networks (wireless and wired)
- Production of ICT
- Consumer devices (televisions, computers, mobile phones)
- Data centres



<https://www.akcp.com/blog/the-real-amount-of-energy-a-data-center-use/>

Amazon Revenue Breakdown



80
Billions \$

Cloud Computing as
an important source
of revenue



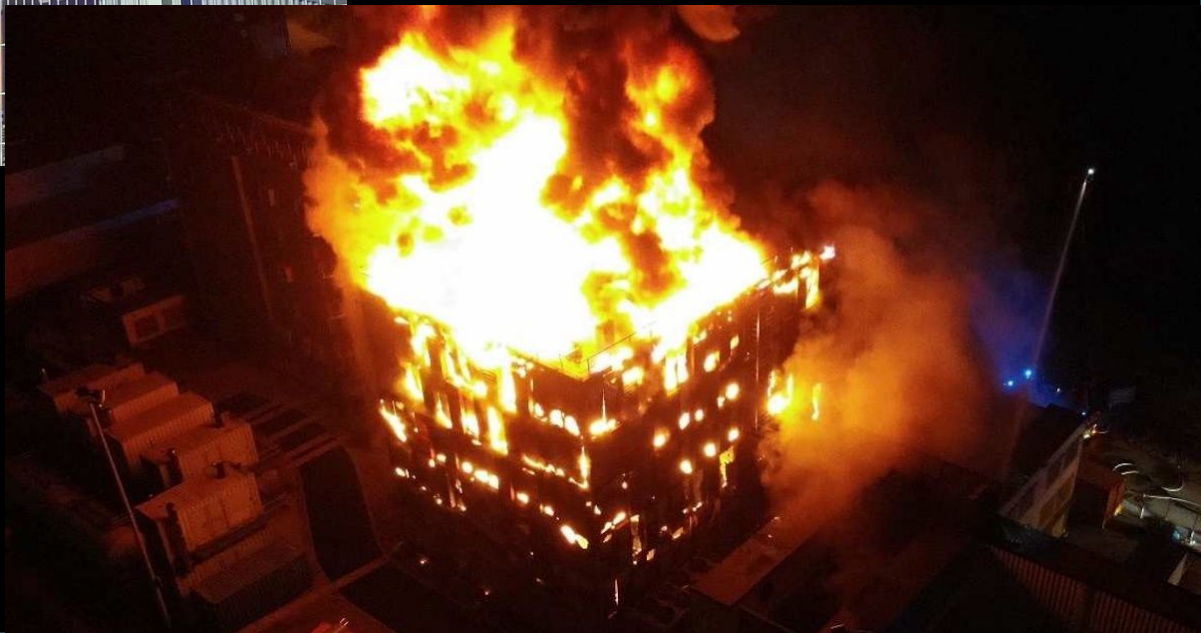
Source:

<https://web.developpez.com/actu/318808/Un-incendie-pourrait-couter-105-millions-d-euros-a-OVHcloud-selon-les-documents-relatifs-a-l-introduction-en-bourse/>

Source:

<https://www.solutions-numeriques.com/exclusif-les-causes-de-lincendie-ovh-strasbourg-des-exlications-stupefiantes/>

3,6 millions de sites internet affectés



Section 2

Words that should be understood after this Curriculum

Notions à Apprendre

IAAS

Architecture

Backend

Broker

Docker

Scaleway

minikube

Virtualisation

SAAS

Kubernetes

PAAS

Kafka

API

AWS

GCP

OVH

Frontend

Cloud

Vagrant

Conteneurisation

Docker
Compose

Azure

Section 3

Cloud Computing



**What do you
understand by the
term “Cloud” ?**



Who coined the term “Cloud Computing” ?

What's interesting [now] is that there is an **emergent new model**, and you all are here because you are part of that new model. I don't think people have really understood how big this opportunity really is. It starts with the premise that the **data services** and **architecture** should be on servers. We call it **cloud computing** – they should be in a “**cloud**” somewhere. And that if you have the right kind of browser or the right kind of access, it doesn't matter whether you have a PC or a Mac or a mobile phone or a BlackBerry or what have you – or new devices still to be developed – you can get access to the cloud. There are a number of companies that have benefited from that. ...

Eric Schmidt (CEO of Google) - 2006



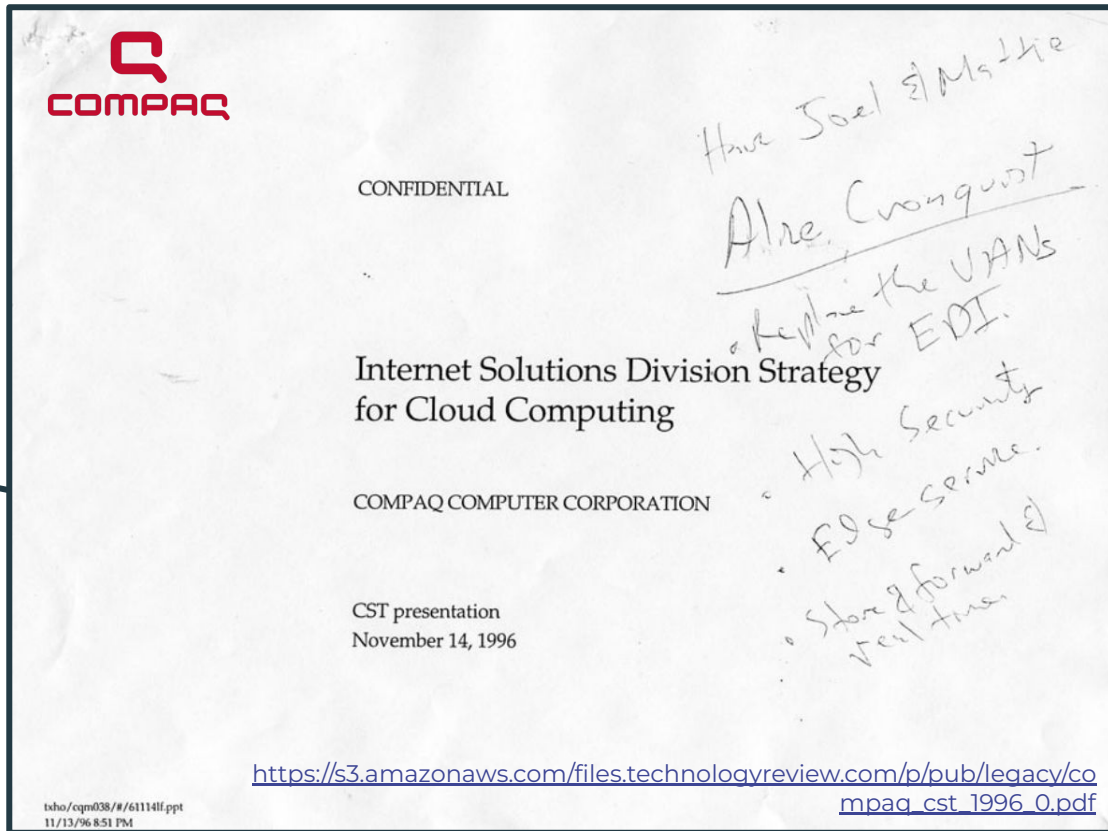
The screenshot shows a Google Press Center page. At the top, there's a search bar and navigation links for 'About Google', 'News from Google', 'Podium Archive', and 'Search Engine Strategies Conference'. The main content is dated 'August 9, 2006' and titled 'Search Engine Strategies Conference' and 'Conversation with Eric Schmidt hosted by Danny Sullivan'. The transcript shows a dialogue between Danny and Eric. Danny starts by welcoming everyone and mentioning the Google Dance. Eric thanks him. Danny then introduces the topic of cloud computing, mentioning the 'cloud' and the fact that data services and architecture should be on servers. Eric responds by saying that this is a historic context to start with, mentioning that 20 years ago everyone was in love with a concept called client/server computing. He then explains that the business model was largely invented by Oracle, which was a direct sales force that would go in and sell complicated software to enterprises that they would integrate and do important business functions. Finally, Eric explains that what's interesting [now] is that there is an emergent new model, and you all are here because you are part of that new model. He repeats the premise that data services and architecture should be on servers, and that if you have the right kind of browser or the right kind of access, it doesn't matter whether you have a PC or a Mac or a mobile phone or a BlackBerry or what have you – or new devices still to be developed – you can get access to the cloud. He mentions that there are a number of companies that have benefited from that, specifically naming Google, Yahoo!, eBay, Amazon, and Linux. The page ends with a URL: <https://www.google.com/press/podium/ses2006.html>.

Who coined the term “Cloud Computing” ?

George Favaloro holds Compaq's 1996 Business Plan. This is the earliest printed document that mentions the term **“cloud computing”**.



<https://prog.world/who-coined-the-term-cloud-computing/>



Cloud - Definitions

Cloud computing is the **on-demand availability of computer system resources**, especially data storage (cloud storage) and computing power, without direct active management by the user.

https://en.wikipedia.org/wiki/Cloud_computing

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(4) **Cloud.** a network of **servers** (= computers that control or supply information to other computers) on which **data** and **software** can be stored or managed and to which **users have access over the internet**

<https://www.oxfordlearnersdictionaries.com/>

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Le **cloud computing** (en français, « **informatique dans les nuages** ») fait référence à l'utilisation de la mémoire et des capacités de calcul **des ordinateurs et des serveurs répartis dans le monde entier et liés par un réseau.**

<https://www.cnil.fr/fr/definition/cloud-computing>

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Cloud computing is the increasingly common practice of providing **'on-demand'** cloud services as IT infrastructure and resources.

<https://www.ovhcloud.com/en/public-cloud/cloud-computing/>

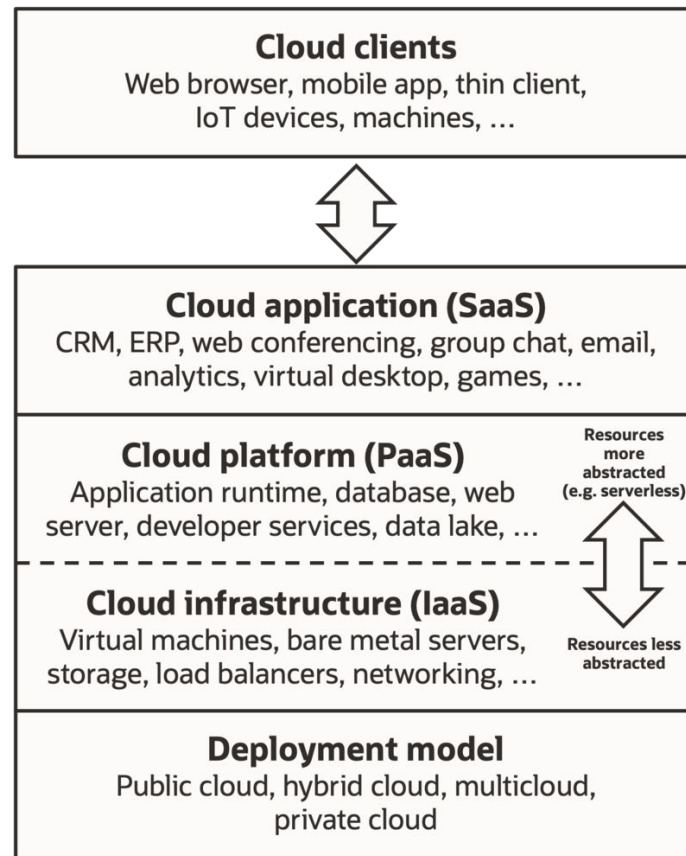
Cloud - Definitions

Cloud computing is a model for enabling **ubiquitous, convenient, on-demand** network access to a **shared pool of configurable computing resources** (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with **minimal management effort** or service provider interaction.

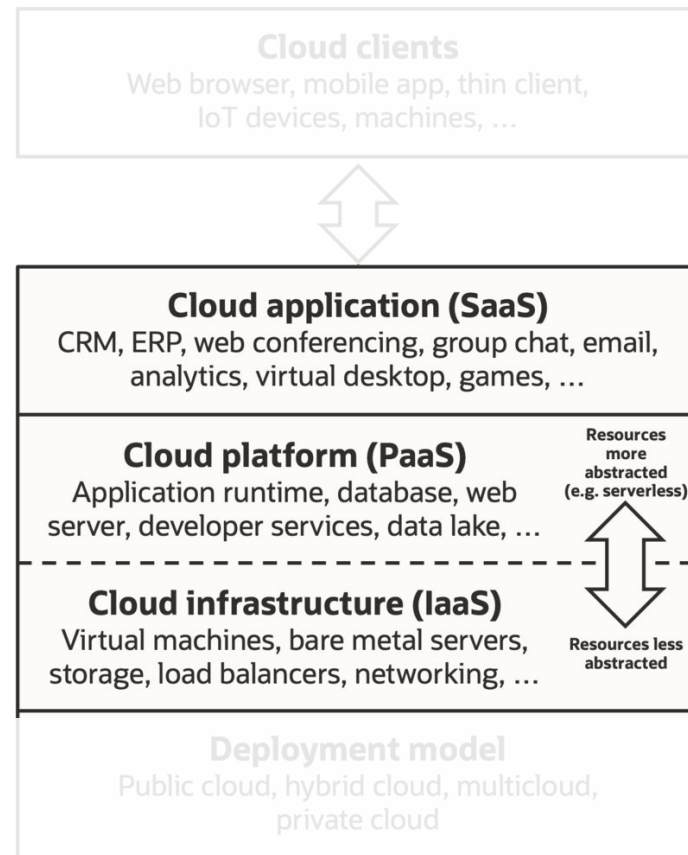
<https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf>

(NIST - National Institute of Standards and Technology - 2011)

Cloud - Service Models



Cloud - Service Models

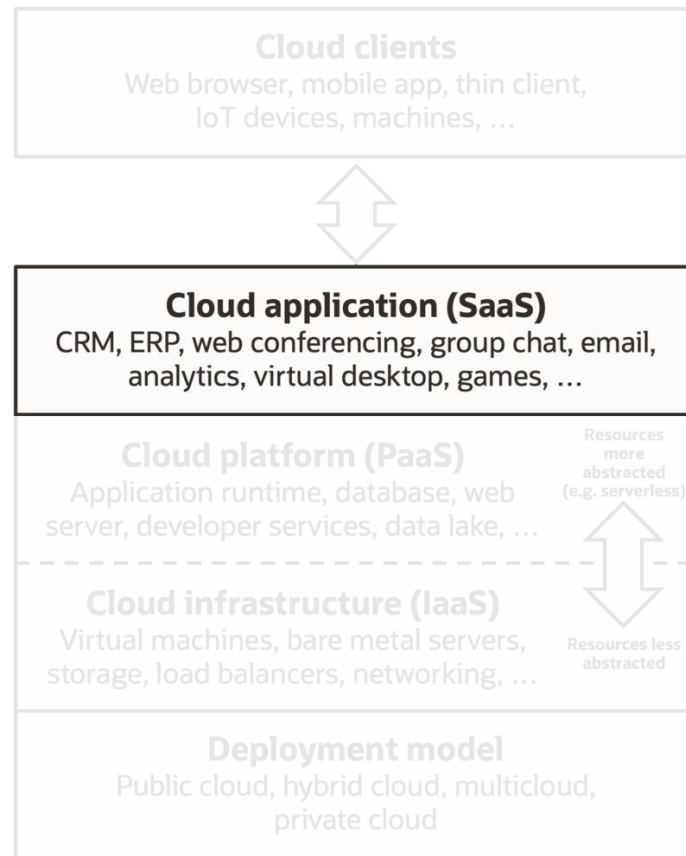


Cloud - Provided Services

Software as as Service (SaaS)

The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure . The applications are accessible from various client devices [...]. The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user specific application configuration settings.

Examples: Gmail, Slack, Google Drive, etc.



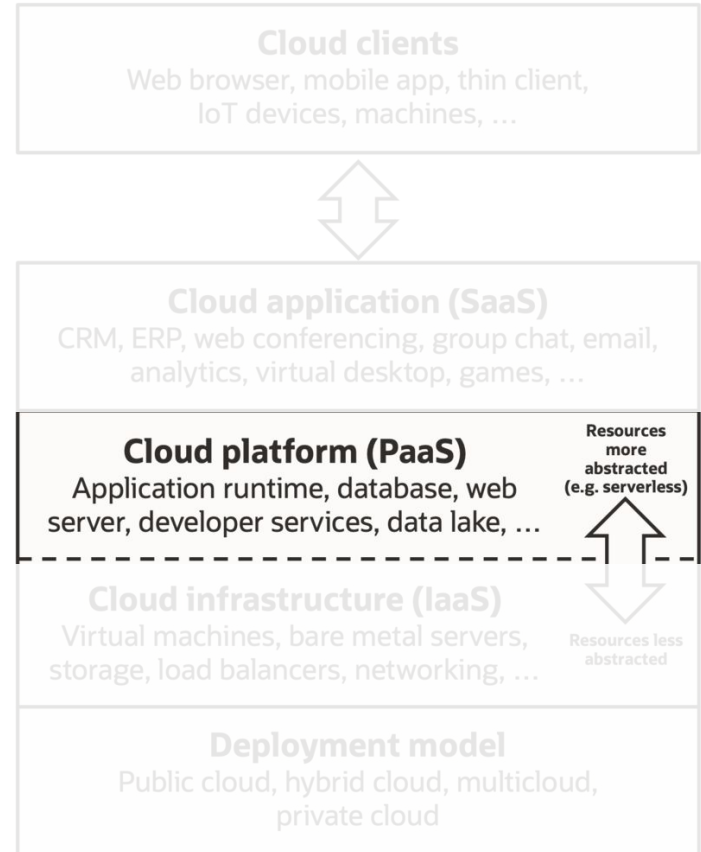
Cloud - Provided Services

Platform as a Service (PaaS)

Platform as a Service (PaaS). The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider. The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly configuration settings for the application-hosting environment.

Examples: Heroku, Google App Engine, Amazon Lambda (FaaS - Function as a Service), ...

<https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf>



https://en.wikipedia.org/wiki/Cloud_computing

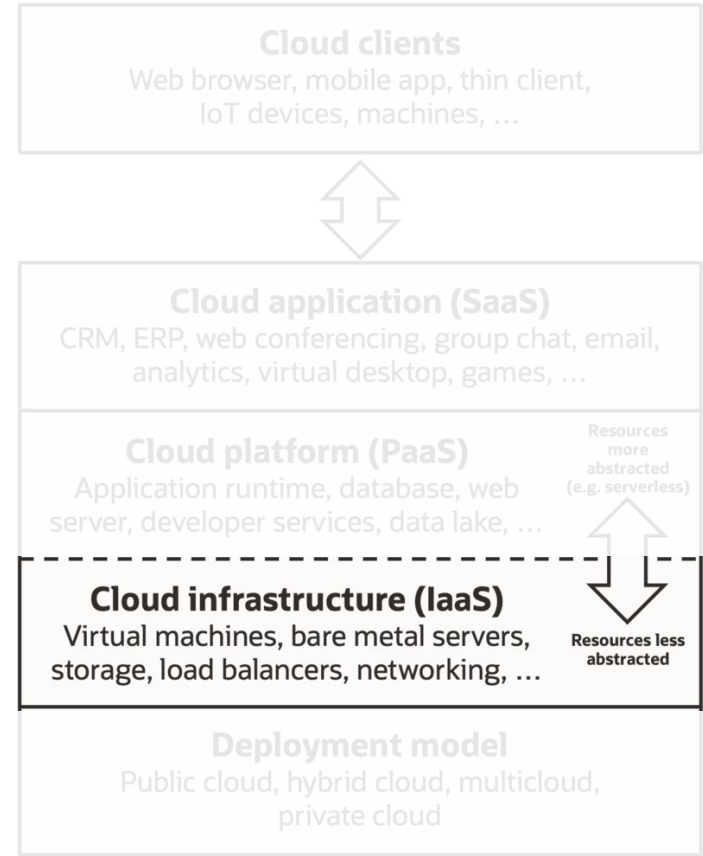
Cloud - Provided Services

Infrastructure as as Service (IaaS)

The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications. The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, and deployed applications; and possibly limited control of select networking components (e.g., host firewalls).

Examples: Google Cloud Platform (GCP), Amazon Web Services (AWS), Google Azure, OVH Public Cloud, ...

<https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf>



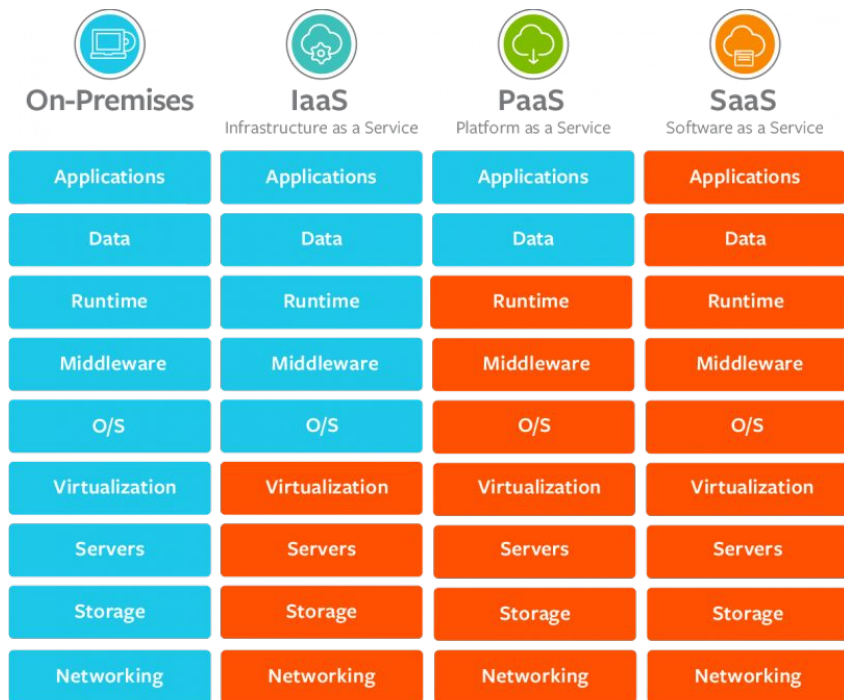
https://en.wikipedia.org/wiki/Cloud_computing

Cloud - Cloud Computing vs On-Premise

On-Premise

On-premises refers to IT infrastructure hardware and software applications that are **hosted on-site**. This contrasts with IT assets that are hosted by a public cloud platform or remote data center.

https://www.insight.com/en_US/content-and-resources/glossary/o/on-premises.html



You Manage

Other Manages

<https://www.bmc.com/blogs/saas-vs-paas-vs-iaas-whats-the-difference-and-how-to-choose/>

Cloud - Deployment Models



Public clouds

A cloud environment created from resources not owned by the end user that can be redistributed to other tenants.



Private clouds

Loosely defined as a cloud environment solely dedicated to the end user, usually within the user's firewall and sometimes on premise.



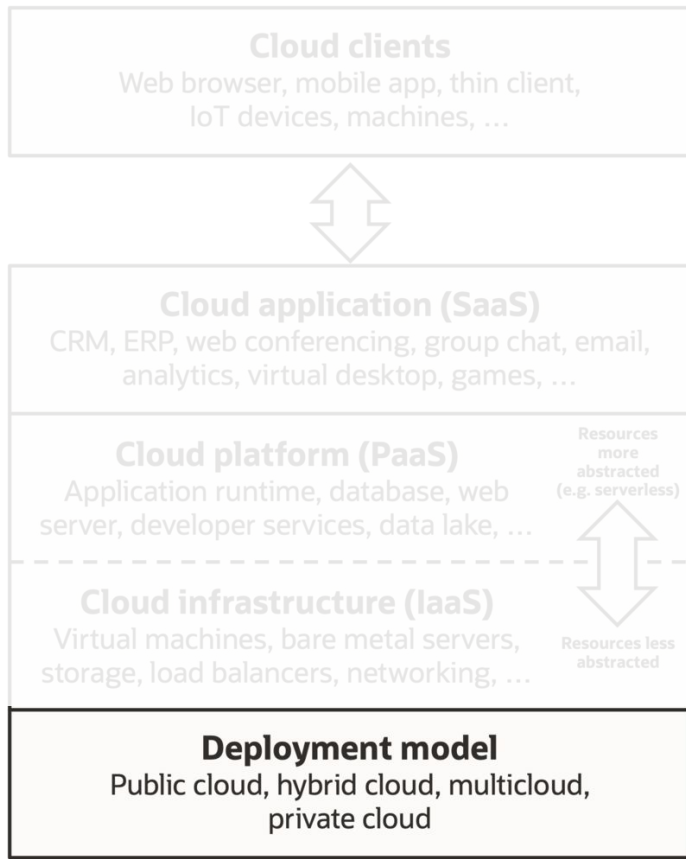
Hybrid clouds

Multiple cloud environments with some degree of workload portability, orchestration, and management among them.



Multiclouds

An IT system that includes more than 1 cloud—public or private—that may or may not be networked together.



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Cloud - Deployment Models

Public Clouds

A public cloud is a pool of virtual resources—developed from hardware owned and managed by a third-party company—that is automatically provisioned and allocated among multiple clients through a self-service interface.

Examples: Google Cloud Platform (GCP), Amazon Web Services (AWS), Google Azure, OVH Public Cloud, ...



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An IT system that includes more than 1 cloud—public or private—that may or may not be networked together.

Cloud - Deployment Models

Private Clouds

Private Cloud are loosely defined as cloud environments solely dedicated to a single end user or group.

Private clouds no longer have to be sourced from **on-premise IT infrastructure**.

Organizations are now building private clouds on rented, vendor-owned data centers located off-premises. This has also led to a number of private cloud subtypes, including **managed private clouds** and **dedicated clouds**.

Examples: OVH Hosted Private Cloud.



Public clouds

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Multiple cloud environments with some degree of workload portability, orchestration, and management among them.



Multiclouds

An IT system that includes more than 1 cloud—public or private—that may or may not be networked together.

Cloud - Deployment Models

Multiclouds and Hybrid Clouds

Multiclouds are a cloud approach made up of more than 1 cloud service, from **more than 1 cloud vendor** - public or private. All hybrid clouds are multiclouds, but not all multiclouds are hybrid clouds. Multiclouds become **hybrid clouds** when **multiple clouds are connected by some form of integration or orchestration**.



Public clouds

A cloud environment created from resources not owned by the end user that can be redistributed to other tenants.



Private clouds

Loosely defined as a cloud environment solely dedicated to the end user, usually within the user's firewall and sometimes on premise.



Hybrid clouds

Multiple cloud environments with some degree of workload portability, orchestration, and management among them.



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Section 3

Cloud and Data Engineering

Cloud & Data Engineering

Cloud & Data Engineering

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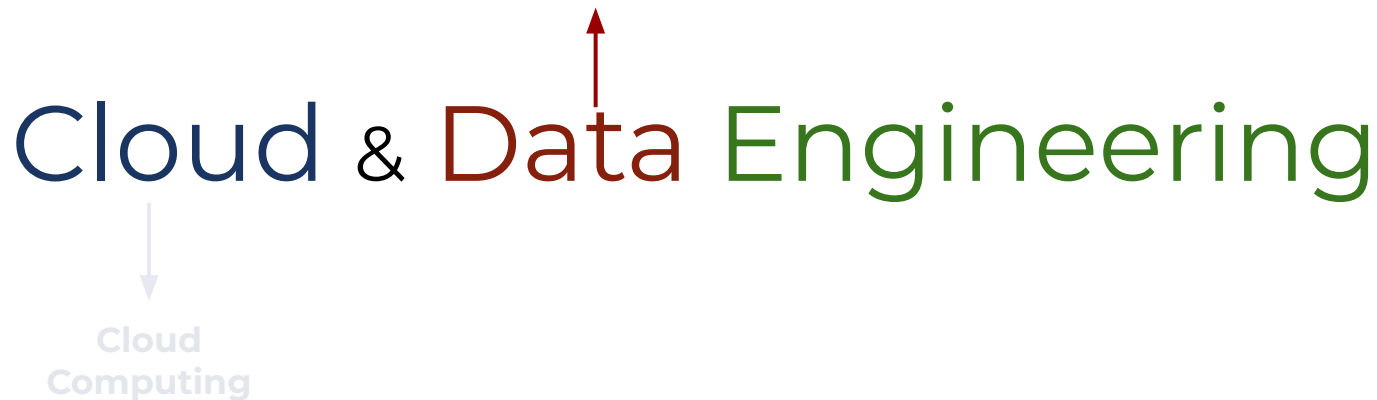
Cloud & Data Engineering



**Cloud
Computing**

Cloud & Data Engineering

Data is the physical representation of information in a manner suitable for communication, interpretation, or processing by human beings or by **automatic means**.



Cloud & Data Engineering

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Cloud & Data Engineering

```
graph TD; CC[Cloud Computing] --> CDE[Cloud & Data Engineering]; DE[Data Engineering] --> CDE;
```

Cloud
Computing

* The **creative application of scientific principles** to **design** or **develop structures**, machines, apparatus, or manufacturing processes, or works utilizing them singly or in combination; or to construct or operate the same with full cognizance of their design; or to forecast their behavior under specific operating conditions; all as respects an intended function, economics of operation and safety to life and property.

Cloud & Data Engineering

Cloud Engineering

Cloud engineering is the application of **engineering disciplines** to **cloud computing**. It brings a systematic approach to concerns of commercialization, standardization, and governance of cloud computing applications. In practice, it leverages the methods and tools of engineering in **conceiving, developing, operating** and **maintaining cloud computing systems** and **solutions**.

https://en.wikipedia.org/wiki/Cloud_engineering

Data Engineering

Data engineering refers to the building of systems to enable the **collection** and **usage** of **data**. This data is usually used to enable subsequent analysis and data science; which often involves machine learning. Making the data usable usually involves substantial **compute** and **storage**, as well as **data processing**.

https://en.wikipedia.org/wiki/Data_engineering

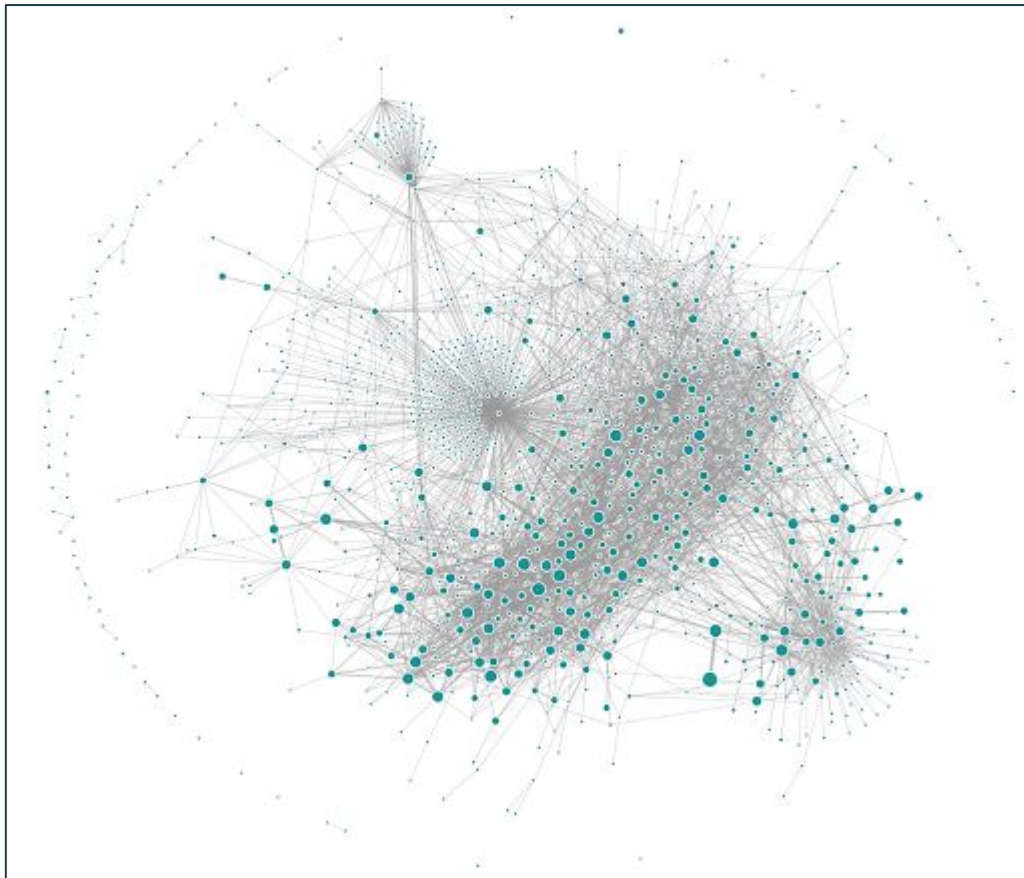
Section 4

Cloud and Architecture - Part I

Uber's microservice architecture circa mid-2018 from Jaeger*

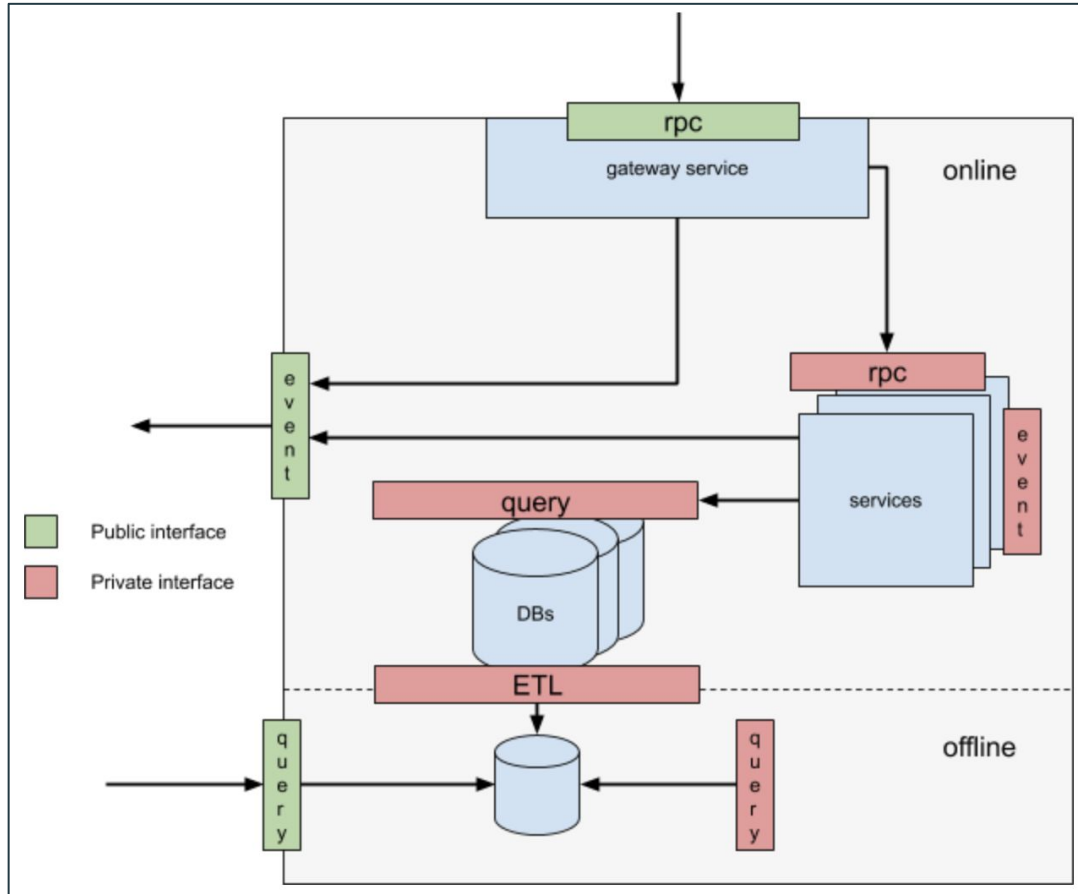


***Jaeger** is open source software for
tracing transactions between
distributed **services**



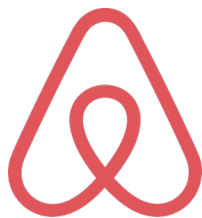
<https://www.uber.com/en-FR/blog/microservice-architecture> [2020]

Gateway API in a Domain-Oriented Microservice Architecture (Uber)

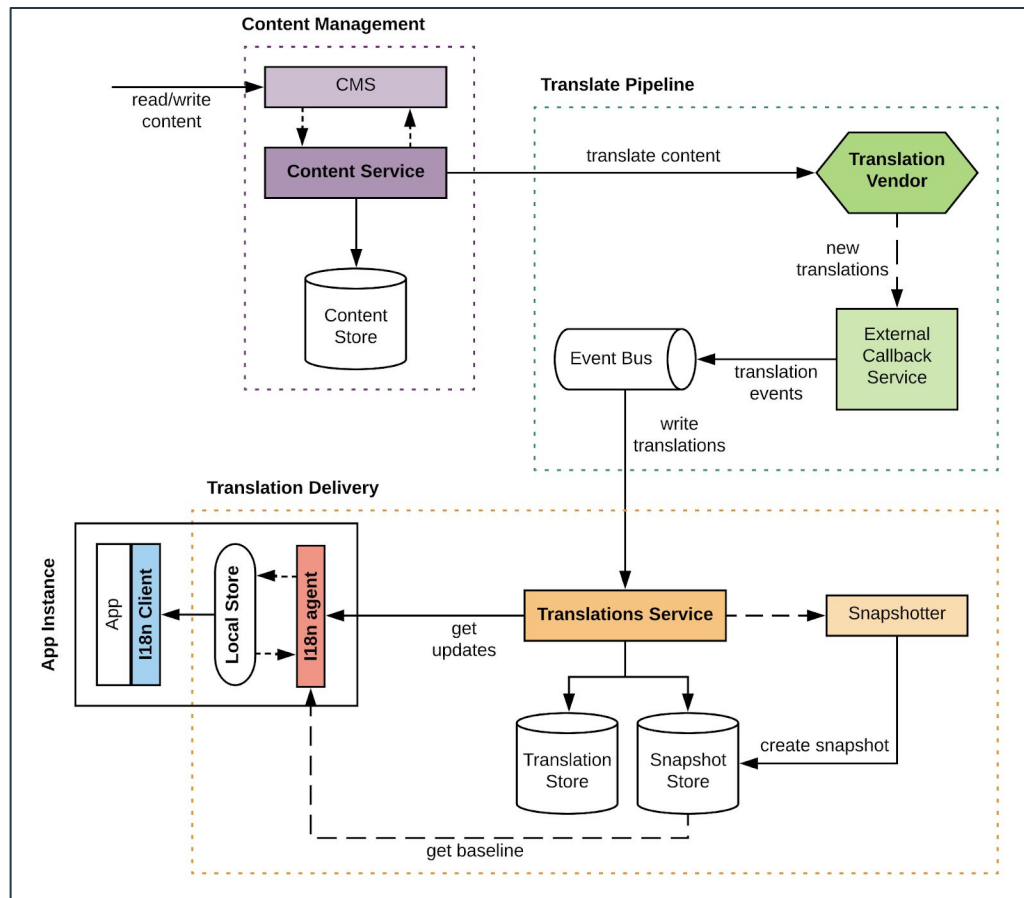


<https://www.uber.com/en-FR/blog/microservice-architecture> [2020]

Airbnb's Internationalization* Platform

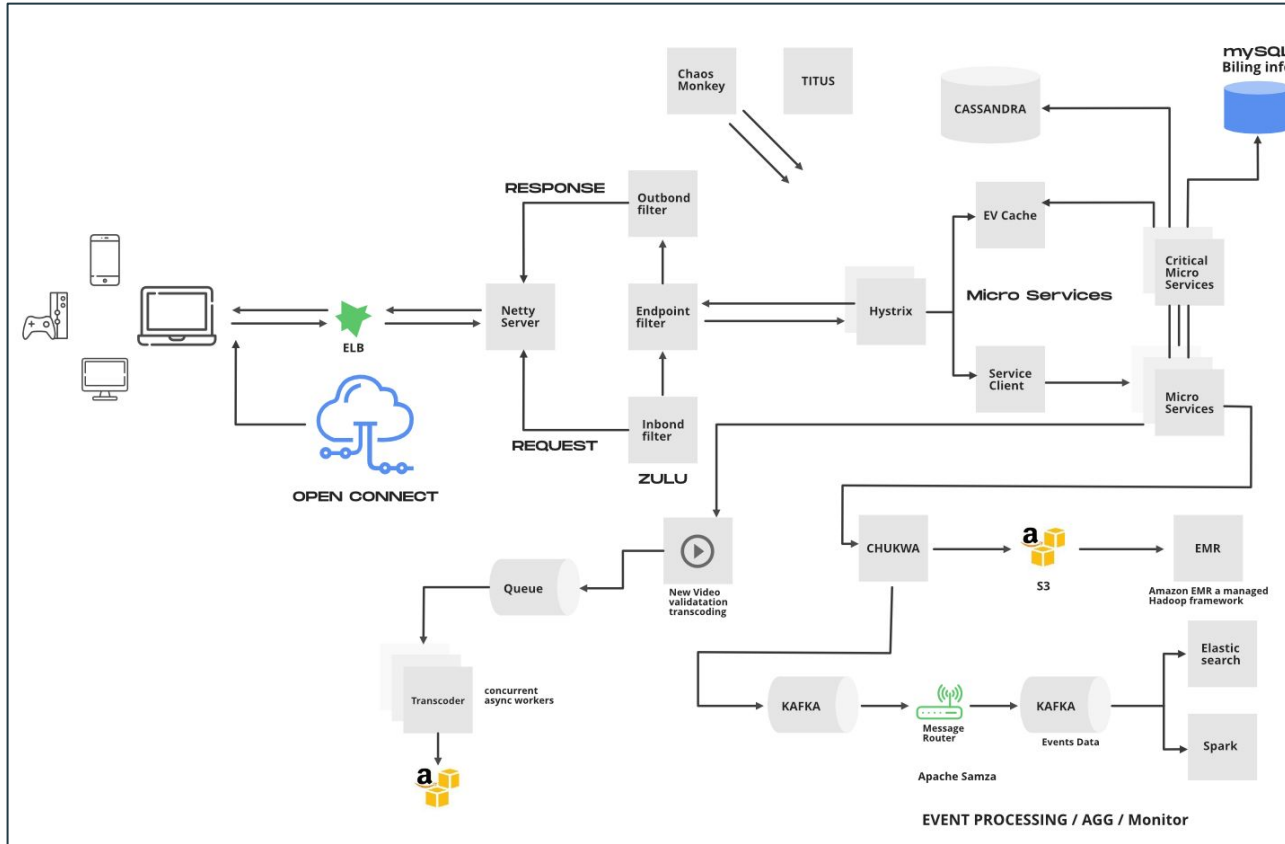


****Internationalization** is the process of adapting software to accommodate for different languages, cultures, and regions (i.e. locales), while minimizing additional engineering changes required for localization.*

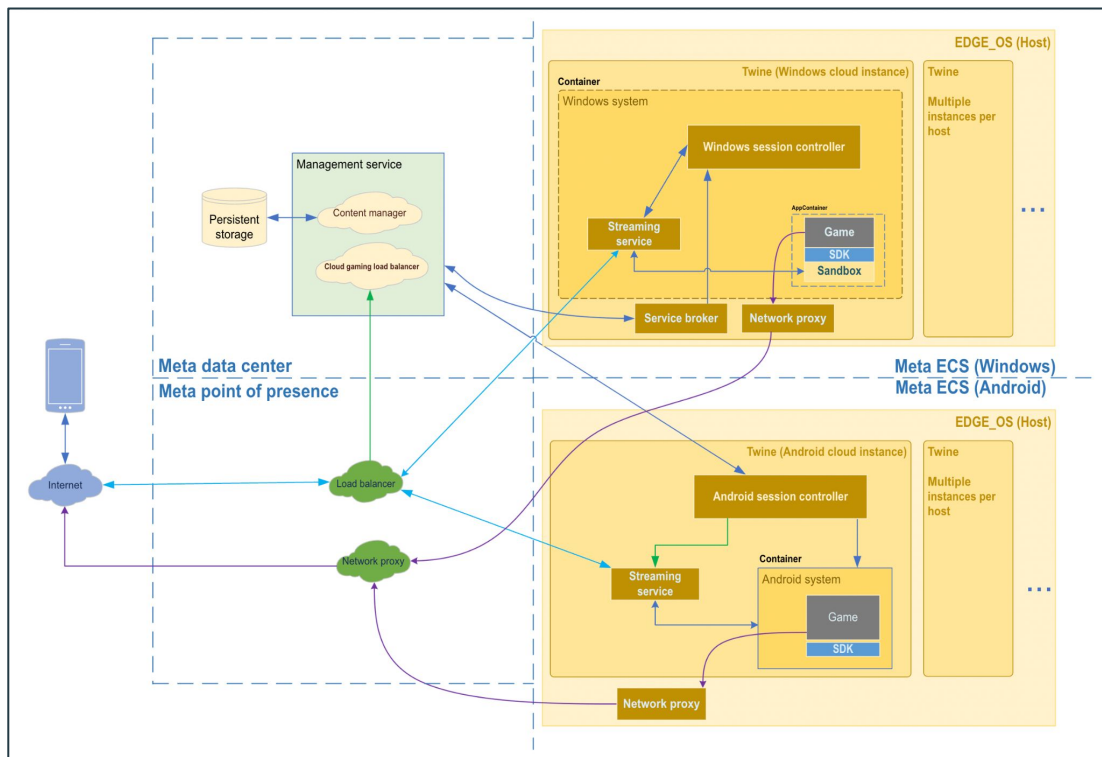
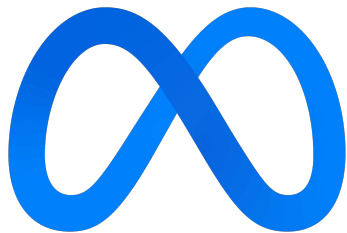


Netflix Backend Architecture

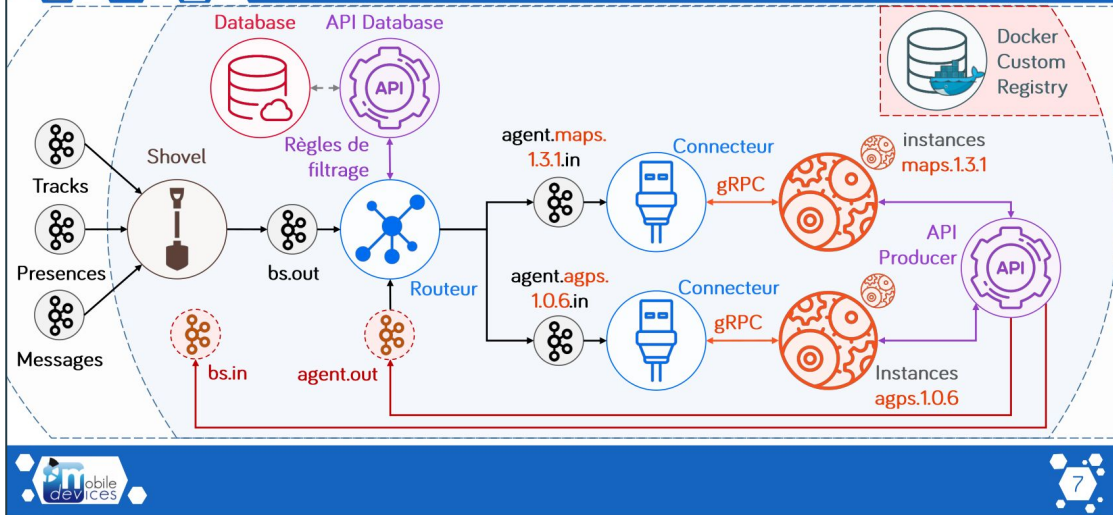
NETFLIX



Meta's Cloud Gaming Infrastructure



Architecture proposée alpha



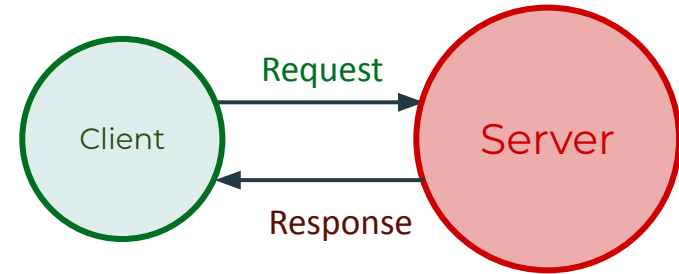
Munic - cloud V4 Draft (2016)



Architecture Patterns - Building Patterns

Client-Server (request-response) Pattern

In the client-server architecture patterns, there are two main components: The client, which is the **service requester**, and the server, which is the **service provider**. Although both client and server may be located within the same system, they often communicate over a network on separate hardware.

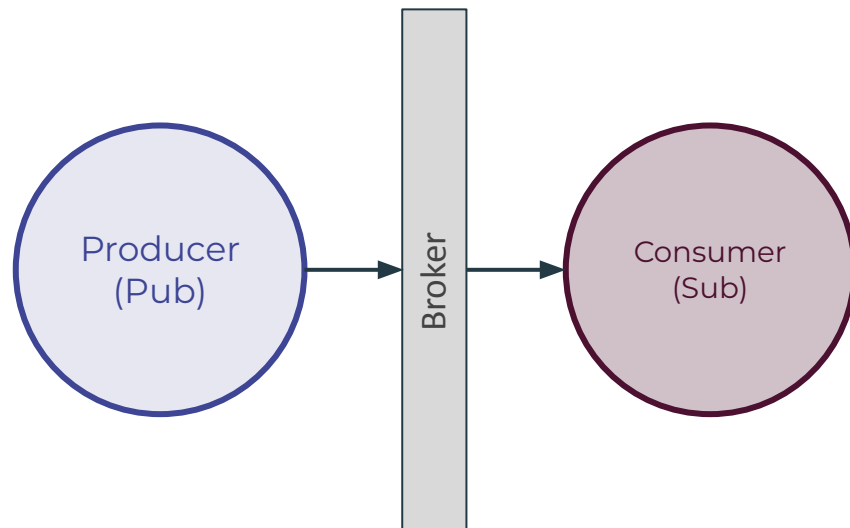


Architecture Patterns - Building Patterns

Publisher-Subscriber (Producer-Consumer) Pattern

Enable an application to **announce events** to multiple interested consumers **asynchronously***, **without coupling** the **senders (publishers)** to the **receivers (consumers)**.

* **Asynchronous communication** differs from **synchronous communication** in the way that in asynchronous communication the producer **does not need to wait** the receiver to handle the produced events/data.



<https://learn.microsoft.com/en-us/azure/architecture/patterns/publisher-subscriber>

Section 5 - Exercise

Cloud and Architecture - Part II -
Implementing a Client-Server Pattern

Architecture Patterns - Client-Server Pattern

Exercise 1:

Implement a simple **REST API** to manage products in a store. A **Product** has a **name**, **price** and **quantities** for attributes. At Least **POST** and **GET** should be implemented.



Your REST API in Python or any other chosen language

cURL linux command line tool

Architecture Patterns - Client-Server Pattern

Exercise 1:

Implement a simple **REST API** to manage products in a store. A **Product** has a **name**, **price** and **quantities** for attributes. At Least **POST** and **GET** should be implemented.

Questions:

What is an API ? Why API and not just a Web Interface ? What a REST API ?

Technologies:

Use whatever language you want. If you are using Python, we suggest **fastapi** and **pydantic**.

Warning:

For now, the data stored by your API does not need to be persistent.

```
curl -X 'POST' 'http://localhost:8000/products/' -H 'accept: application/json' -H 'Content-Type: application/json' -d '{ "name": "Orange", "price": 4, "quantity": 12 }'
```

```
curl -X 'GET' 'http://localhost:8000/products/Orange'
```

```
{ "name": "Orange", "price": 4, "quantity": 12 }
```

Your REST API may also be implemented in any other chosen language

CURL linux command line tool